

```
// Contains a setting entry (or entries, in the case of
collisions).
message Entry {
  string key = 1;
  string value = 2;
}

// List of setting entries - more than one implies a state
key collision
repeated Entry entries = 1;
}
```

sawtooth.settings.vote.proposals is a base64 encoded string of the protobuf message *SettingCandidates*. This setting cannot be modified by a proposal or a vote. The setting is stored as defined by the following protocol buffers definition. The value returned by this setting is a base64 encoded *SettingCandidates* message:

```
// Contains the vote counts for a given proposal.
message SettingCandidate {
  // An individual vote record
  message VoteRecord {
    // The public key of the voter
    string public_key = 1;

    // The voter's actual vote
    SettingVote.Vote vote = 2;
  }

  // The proposal id, a hash of the original proposal
  string proposal_id = 1;

  // The active proposal
  SettingProposal proposal = 2;

  // list of votes
  repeated VoteRecord votes = 3;
}

// Contains all the setting candidates up for vote.
message SettingCandidates {
  repeated SettingCandidate candidates = 1;
}
```

Addressing

When a setting is read or changed, it is accessed by addressing it using the following algorithm:

- Setting keys are broken into four parts, based on the dots in the string. For example, the address for the key a.b.c is computed based on a, b, c and the empty string.
- A longer key, for example a.b.c.d.e, is still broken into

four parts, but the remaining pieces are in the last part: a, b, c and d.e.

- Each of these pieces has a short hash computed (the first 16 characters of its SHA256 hash in hex) and is joined into a single address, with the settings name space (000000) added at the beginning.

For example, the setting *sawtooth.settings.vote.proposals* could be set like this:

```
>>> '000000' + hashlib.sha256('sawtooth'.encode()).
hexdigest()[:16] + \
hashlib.sha256('config'.encode()).hexdigest()[:16] + \
hashlib.sha256('vote'.encode()).hexdigest()[:16] + \
hashlib.sha256('proposals'.encode()).hexdigest()[:16]
'000000a87cb5eafdc6a 8b79606fb3afea5bdab27447
4a6aa82c1c0cbf0fbcaf64c0b'
```

Transaction header

Inputs and outputs: The inputs for the config family transactions must include:

- The address of *sawtooth.settings.vote.proposals*
- The address of *sawtooth.settings.vote.authorized_keys*
- The address of *sawtooth.settings.vote.approval_threshold*
- The address of the setting being changed

The outputs for the config family transactions must include:

- The address of *sawtooth.settings.vote.proposals*
- The address of the setting being changed

Dependencies: None.

Family:

- family_name*: sawtooth_settings
- family_version*: 1.0

Transaction payload

Setting transaction family payloads are defined by the following protocol buffers code:

```
""" Setting Payload
// - Contains either a proposal or a vote.
message SettingPayload {
  // The action indicates data is contained within this
  payload
  enum Action {
    // A proposal action - data will be a SettingProposal
    PROPOSE = 0;
    // A vote action - data will be a SettingVote
    VOTE = 1;
  }

  // The action of this payload
  Action action = 1;
```